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CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY AND ENGINEERING

DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY AND RESEARCH

DEPARTMENT OF COMPUTER ENGINEERING

A.Y 2026 – 27 Semester – 3

LAB MANUAL

CSUC201 Fundamentals of Data Structures and Algorithms

DEPSTAR

Practical 9: Graph Traversal: Depth-First Search and Breadth-First Search

Practical Aim :

To implement binary tree traversals and binary search tree insertion, and develop the ability to reason about how the order of visiting nodes changes the information retrieved from a tree..

Problem 1 :

A disaster relief team needs to inspect all buildings in a city after an earthquake. The city is

mapped as a network where buildings are connected by roads. One team starts at a building

and goes as deep into each road as possible before backtracking, while a second team spreads

out level by level — first checking all buildings directly connected to the start, then all buildings

two roads away, and so on. Given the city map as a graph and a starting building, implement

both exploration strategies and print the order in which buildings are visited by each team.

Describe how the order of buildings visited differs between the two strategies on the same

graph. What data structure does each strategy rely on to decide which building to visit next, and

how does that structure cause the difference in order?

For Code:

https://github.com/KaushalPatel22/FDSA/tree/main/FDSAGITSUBMITION

Output:

```
PS C:\Users\kaush\OneDrive\Desktop\code\DSA> cd "c:\Users\kaush\OneDrive\Desktop\code\DSA\" ; 
d_BFS }
Enter number of nodes and edges: 6 6
Enter edges:
0 1
0 2
1 3
1 4
2 5
4 5
Enter starting node: 0
DFS: 0 1 3 4 5 2
BFS: 0 1 2 3 4 5
PS C:\Users\kaush\OneDrive\Desktop\code\DSA>
```

Outcome/Conclusion:

Learned to implement DFS using recursion and BFS using a queue to traverse a graph.

Problem: Faced:

Understanding the difference between DFS and BFS and managing visited nodes correctly.

Key Questions / Analysis / Interpretation

1. Both strategies visit every reachable node exactly once. If the goal is simply to check whether a specific building exists in the network, which strategy would find it sooner if the building is very close to the start? What if it is very far and deep? Explain your reasoning without tracing through any specific example.

Answer:

BFS finds a nearby building sooner because it explores level by level. DFS may find a deep building sooner if it follows the correct path.

2. Your DFS implementation likely uses recursion. What real-world constraint on the computer could cause your recursive DFS to fail on a very large city map, and how would you restructure your solution to avoid that failure?

Answer:

Recursive DFS may cause stack overflow on a very large graph. Using an iterative DFS with an explicit stack can avoid this problem.